

Results

Aug 12, 2024

```
knitr::opts_chunk$set(echo = TRUE)

repo_dir = "/Users/haya1/Documents/LanguageModel_Labels/congressional_bills"
results_dir = file.path(repo_dir, "05_results")

knitr::opts_knit$set(root.dir = results_dir)
setwd(results_dir)

require(dplyr)
require(ggplot2)
require(latex2exp)
```

Proxy on the LHS

```
beta1 <- read.csv(file.path(repo_dir, "04_simulation/simulations_averaged_lhs.csv")) %>%
  filter(coef_name!="(Intercept)") # Plot only \beta_1
```

Figures

```
figures_dir <- file.path(results_dir, "figures")
dir.create(figures_dir, showWarnings=FALSE)

legend_colors <- c("Yhuman_V"="black", "Yllm_V"="#1f77b4", "train_Yhuman_V"="#008000", "test_Ytilde_V"=
legend_labels <- c("Yhuman_V"=TeX(r"($\beta_1^*$)"),
                  "Yllm_V"=TeX(r"($\hat{\beta}_1$ (LLM))"),
                  "train_Yhuman_V"=TeX(r"($\beta_1^{\text{train}}$ (Human))"),
                  "test_Ytilde_V"=TeX(r"($\tilde{\beta}_1$ (Debiased))"))

regression_label <- TeX(r"(LHS Regression: $1\{Y=y\} = \beta_0 + \beta_1 V$)")
```

Figure 1: Density Plot of Bias(β_1, β_1^*) by Training Proportion

```
# Figure 1
p1 <- ggplot(data=beta1, aes(x=bias, color=regression)) +
  geom_vline(xintercept=0, color="black", linetype = "dashed") +
  # geom_histogram(alpha=0, bins = 40, position = "identity") +
  geom_density() +
```

```

labs(title=TeX(r"(Density Plots of $Bias(\beta_1, \beta_1^*)$ by Training Proportion)"),
      subtitle=regression_label) +
xlab(TeX(r"($Bias(\beta_1, \beta_1^*) = E[\beta_1] - \beta_1^*$)")) +
facet_grid(train_proportion ~ ., scales="free_y") +
scale_color_manual(name="", values=legend_colors, breaks=names(legend_colors), labels=legend_labels)
theme_bw() +
theme(panel.grid.major=element_blank(), panel.grid.minor=element_blank(), legend.position="right")
p1

```

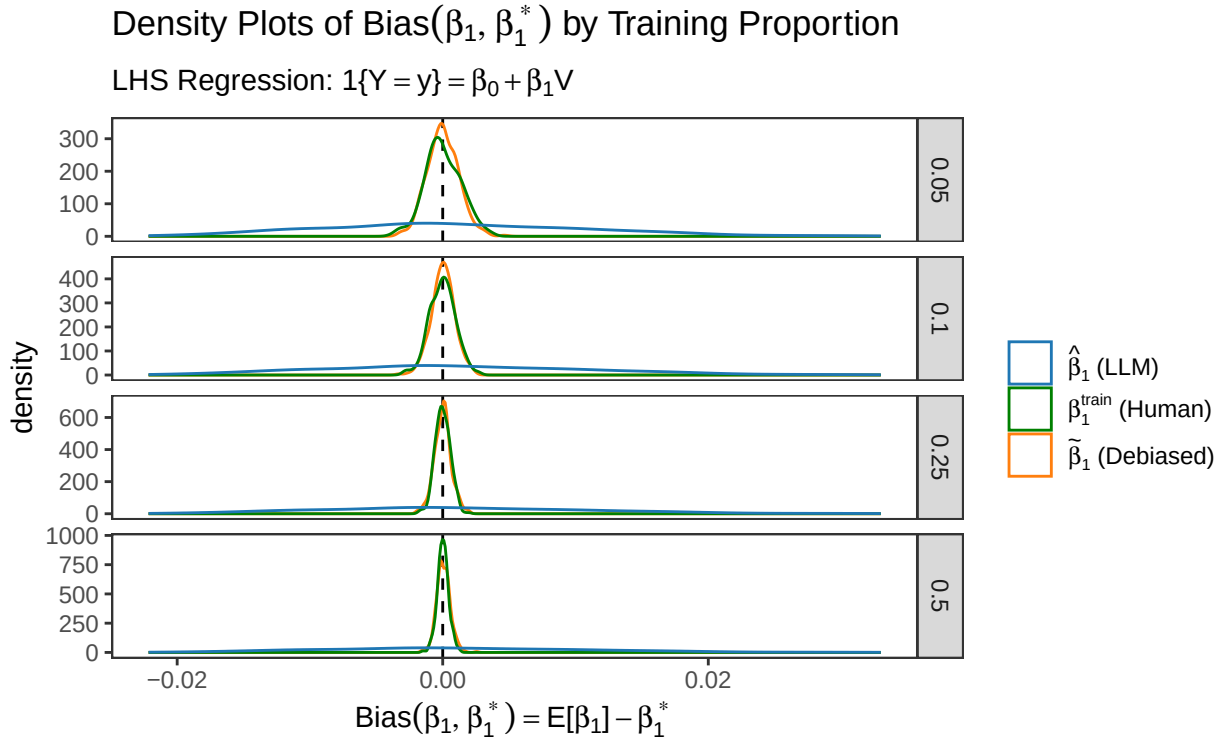


Figure 2: Empirical CDF Plot of $MSE(\beta_1, \beta_1^*)$ by Training Proportion

```

p2 <- ggplot(data=beta1, aes(x=mse, color=regression)) +
  stat_ecdf() +
  labs(title=TeX(r"(Empirical CDF of $MSE(\beta_1, \beta_1^*)$ by Training Proportion)"),
        subtitle=regression_label) +
  facet_grid(train_proportion ~ .) +
  xlab(TeX(r"($MSE(\beta_1, \beta_1^*) = E[(\beta_1 - \beta_1^*)^2]$)")) +
  scale_color_manual(name="", values=legend_colors, labels=legend_labels, breaks=names(legend_colors))
  theme_bw() +
  theme(panel.grid.major=element_blank(), panel.grid.minor=element_blank(), legend.position="right")
p2

```

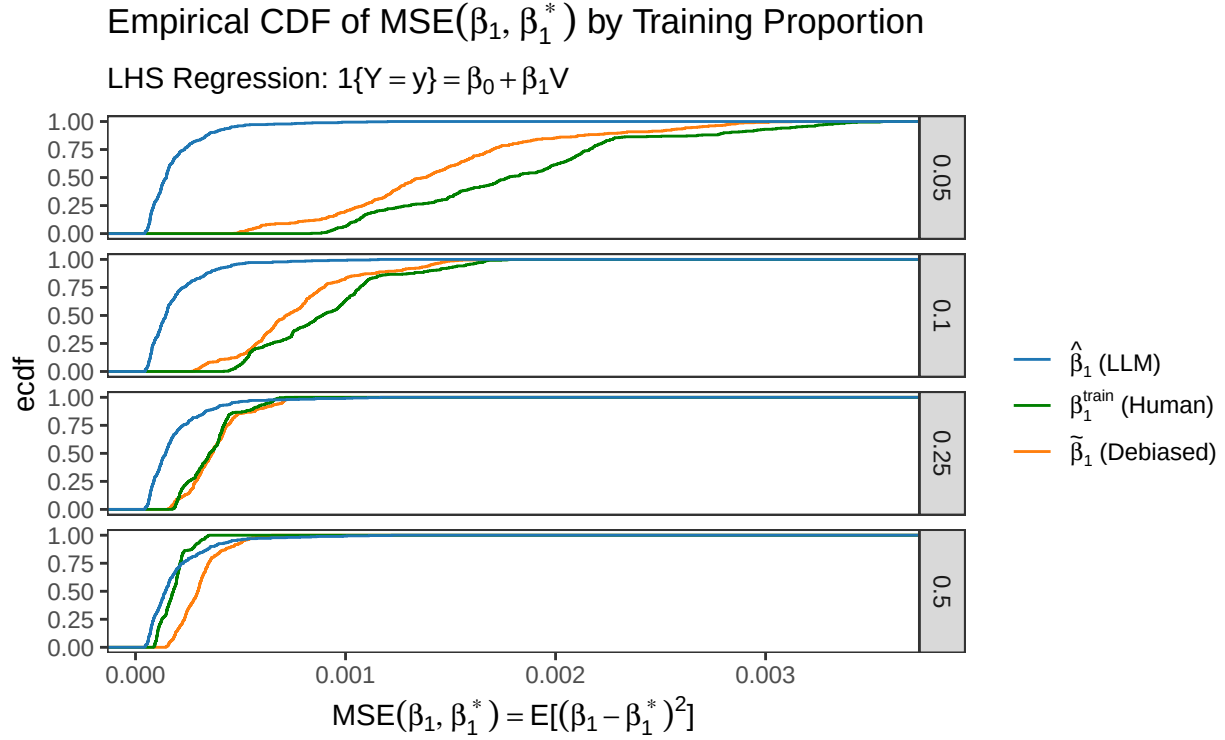


Figure 3: Density Plot of $\text{Bias}(\beta_1, \beta_1^*)$ by Model and Training Proportion

```
p3 <- ggplot(data=beta1, aes(x=bias, color=regression)) +
  geom_vline(xintercept=0, color="black", linetype = "dashed") +
  # geom_histogram(alpha=0, bins = 40, position = "identity") +
  geom_density() +
  labs(title=TeX(r"(Density Plots of $Bias(\beta_1, \beta_1^*)$ by Model & Training Proportion)"),
        subtitle=regression_label) +
  xlab(TeX(r"($Bias(\beta_1, \beta_1^*) = E[\beta_1] - \beta_1^*$)")) +
  facet_grid(train_proportion + model ~ ., scales="free_y") +
  scale_color_manual(name="", values=legend_colors, breaks=names(legend_colors), labels=legend_labels) +
  theme_bw() +
  theme(panel.grid.major=element_blank(), panel.grid.minor=element_blank(), legend.position="right")
p3
```

Density Plots of Bias(β_1, β_1^*) by Model & Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$

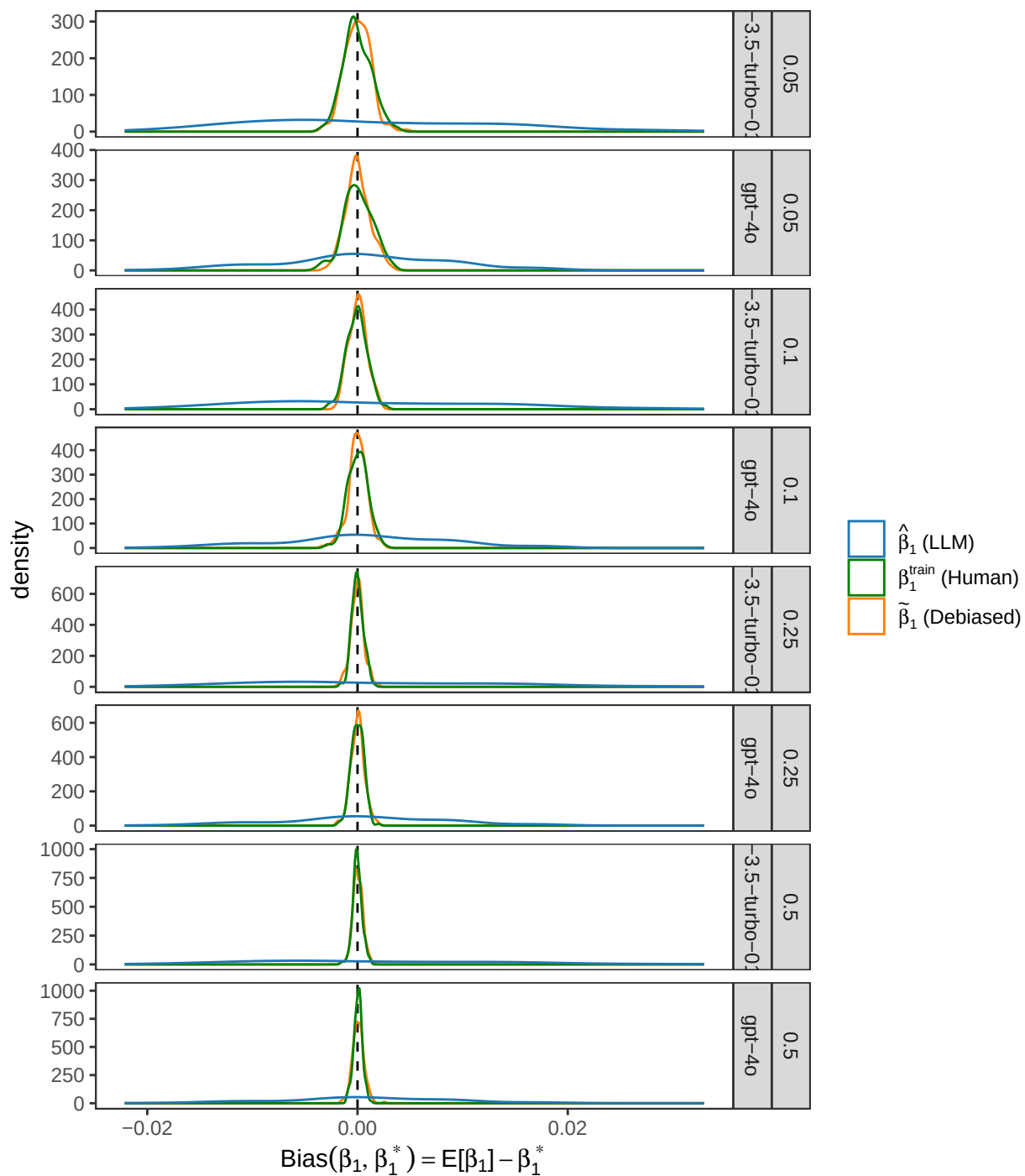
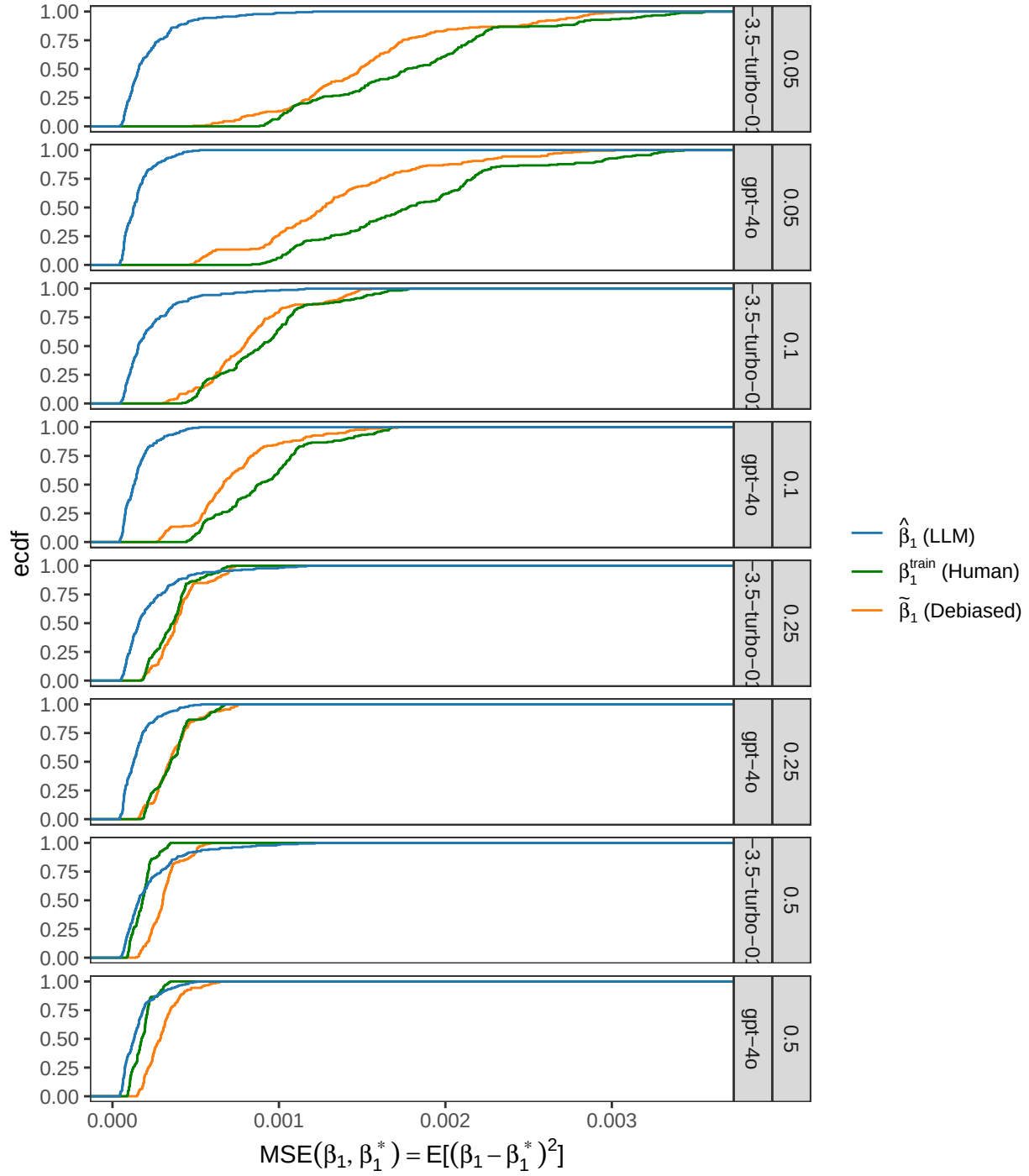


Figure 4: Empirical CDF Plot of $\text{MSE}(\beta_1, \beta_1^*)$ by Model and Training Proportion

```
p4 <- ggplot(data=beta1, aes(x=mse, color=regression)) +
  stat_ecdf() +
  labs(title=TeX(r"(Empirical CDF of  $\text{MSE}(\beta_1, \beta_1^*)$  by Model & Training Proportion)"),
        subtitle=regression_label) +
  xlab(TeX(r"( $\text{MSE}(\beta_1, \beta_1^*) = E[(\beta_1 - \beta_1^*)^2]$ )")) +
  facet_grid(train_proportion + model ~ ., scales="free_y") +
  scale_color_manual(name="", values=legend_colors, labels=legend_labels, breaks=names(legend_colors)) +
  theme_bw() +
  theme(panel.grid.major=element_blank(), panel.grid.minor=element_blank(), legend.position="right")
p4
```

Empirical CDF of $\text{MSE}(\beta_1, \beta_1^*)$ by Model & Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$



Save Figures

```
ggsave(plot=p1, file=file.path(figures_dir, "figure1_lhs.png"), width=6, height=4)
ggsave(plot=p2, file=file.path(figures_dir, "figure2_lhs.png"), width=6, height=4)
ggsave(plot=p3, file=file.path(figures_dir, "figure3_lhs.png"), width=6, height=4*2)
ggsave(plot=p4, file=file.path(figures_dir, "figure4_lhs.png"), width=6, height=4*2)
```

Dataframes/Tables

```
tables_dir <- file.path(results_dir, "tables")
dir.create(figures_dir, showWarnings=FALSE)
```

```
alpha = 0.10 # alpha = 0.05 # TODO: in all simulations, I've been using 95%CI, with 2.5%,97.5% percenti
probs = c(alpha/2, 1-alpha/2)
```

Dataframe 1: Bias Summary Statistics by Training Proportion

```
df1.bias <- beta1 %>%
  mutate(statistic=bias) %>%
  group_by(train_proportion, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df1.bias
```

```
## # A tibble: 12 x 7
##   train_proportion regression      mean      sd    median    p05    p95
##         <dbl> <chr>          <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1         0.05 Yllm_V          0.00112 1.02e-2 2.68e-4 -1.45e-2 1.82e-2
## 2         0.05 test_Ytilde_V 0.0000171 1.19e-3 -1.28e-5 -1.80e-3 1.95e-3
## 3         0.05 train_Yhuman_V 0.0000261 1.36e-3 -8.09e-5 -2.04e-3 2.29e-3
## 4         0.1  Yllm_V          0.00118 1.02e-2 3.11e-4 -1.45e-2 1.85e-2
## 5         0.1  test_Ytilde_V 0.0000172 8.85e-4 4.54e-5 -1.45e-3 1.45e-3
## 6         0.1  train_Yhuman_V -0.0000194 9.70e-4 -7.54e-6 -1.54e-3 1.58e-3
## 7         0.25 Yllm_V          0.00116 1.02e-2 4.46e-4 -1.45e-2 1.85e-2
## 8         0.25 test_Ytilde_V 0.00000245 6.43e-4 2.97e-5 -1.02e-3 1.11e-3
## 9         0.25 train_Yhuman_V 0.00000284 5.73e-4 -1.45e-5 -8.59e-4 9.52e-4
## 10        0.5  Yllm_V          0.00116 1.02e-2 2.94e-4 -1.46e-2 1.82e-2
## 11        0.5  test_Ytilde_V 0.0000597 5.30e-4 4.68e-5 -7.19e-4 9.30e-4
## 12        0.5  train_Yhuman_V 0.0000141 4.24e-4 1.11e-5 -6.70e-4 7.58e-4
```

Dataframe 2: MSE Summary Statistics by Training Proportion

```
df2.mse <- beta1 %>%
  mutate(statistic=mse) %>%
  group_by(train_proportion, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df2.mse
```

```
## # A tibble: 12 x 7
##   train_proportion regression      mean      sd  median      p05      p95
##   <dbl> <chr>          <dbl>    <dbl>   <dbl>    <dbl>    <dbl>
## 1     0.05 Yllm_V      0.000190 0.000160 0.000146 0.0000591 4.54e-4
## 2     0.05 test_Ytilde_V 0.00145 0.000572 0.00138 0.000576 2.67e-3
## 3     0.05 train_Yhuman_V 0.00183 0.000637 0.00177 0.000965 3.14e-3
## 4     0.1  Yllm_V      0.000190 0.000160 0.000144 0.0000592 4.58e-4
## 5     0.1  test_Ytilde_V 0.000768 0.000296 0.000718 0.000321 1.40e-3
## 6     0.1  train_Yhuman_V 0.000909 0.000317 0.000890 0.000493 1.57e-3
## 7     0.25 Yllm_V      0.000190 0.000162 0.000144 0.0000603 4.58e-4
## 8     0.25 test_Ytilde_V 0.000381 0.000138 0.000364 0.000183 6.97e-4
## 9     0.25 train_Yhuman_V 0.000361 0.000127 0.000353 0.000194 6.36e-4
## 10    0.5  Yllm_V      0.000190 0.000162 0.000142 0.0000591 4.57e-4
## 11    0.5  test_Ytilde_V 0.000305 0.000103 0.000297 0.000167 5.11e-4
## 12    0.5  train_Yhuman_V 0.000181 0.0000633 0.000180 0.0000960 3.09e-4
```

Dataframe 3: Coverage Summary Statistics by Training Proportion

```
df3.coverage <- beta1 %>%
  mutate(statistic=coverage) %>%
  group_by(train_proportion, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df3.coverage
```

```
## # A tibble: 12 x 7
##   train_proportion regression      mean      sd  median      p05      p95
##   <dbl> <chr>          <dbl>    <dbl>   <dbl>    <dbl>    <dbl>
## 1     0.05 Yllm_V      0.810 0.162    0.890 0.476 0.95
## 2     0.05 test_Ytilde_V 0.937 0.00827 0.938 0.923 0.949
## 3     0.05 train_Yhuman_V 0.945 0.00735 0.945 0.933 0.958
## 4     0.1  Yllm_V      0.809 0.164    0.889 0.463 0.950
## 5     0.1  test_Ytilde_V 0.944 0.00746 0.944 0.931 0.956
```

```
## 6          0.1  train_Yhuman_V 0.948 0.00749  0.948 0.935 0.961
## 7          0.25 Yllm_V          0.810 0.164   0.887 0.467 0.952
## 8          0.25 test_Ytilde_V  0.948 0.00707  0.948 0.936 0.958
## 9          0.25 train_Yhuman_V 0.950 0.00715  0.95  0.936 0.961
## 10         0.5  Yllm_V          0.810 0.164   0.889 0.455 0.953
## 11         0.5  test_Ytilde_V  0.948 0.00665  0.948 0.936 0.958
## 12         0.5  train_Yhuman_V 0.950 0.00677  0.95  0.938 0.961
```

Dataframe 4: Bias Summary Statistics by Model & Training Proportion

```
df4.bias <- beta1 %>%
  mutate(statistic=bias) %>%
  group_by(train_proportion, model, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df4.bias
```

```
## # A tibble: 24 x 8
##   train_proportion model regression    mean    sd  median    p05    p95
##   <dbl> <chr> <chr>    <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     0.05 gpt-3~ Yllm_V    1.63e-3 1.19e-2 -3.97e-4 -0.0159 0.0228
## 2     0.05 gpt-3~ test_Ytil~ 8.41e-6 1.22e-3 2.89e-5 -0.00181 0.00167
## 3     0.05 gpt-3~ train_Yhu~ -2.35e-5 1.33e-3 -1.62e-4 -0.00204 0.00220
## 4     0.05 gpt-4o Yllm_V    6.13e-4 8.08e-3 4.22e-4 -0.0131 0.0146
## 5     0.05 gpt-4o test_Ytil~ 2.58e-5 1.15e-3 -3.88e-5 -0.00177 0.00216
## 6     0.05 gpt-4o train_Yhu~ 7.58e-5 1.39e-3 4.48e-6 -0.00197 0.00230
## 7     0.1  gpt-3~ Yllm_V    1.71e-3 1.19e-2 -9.76e-5 -0.0160 0.0227
## 8     0.1  gpt-3~ test_Ytil~ 3.81e-5 8.55e-4 4.52e-5 -0.00133 0.00149
## 9     0.1  gpt-3~ train_Yhu~ -7.26e-5 9.79e-4 -4.07e-5 -0.00156 0.00157
## 10    0.1  gpt-4o Yllm_V    6.53e-4 8.15e-3 5.68e-4 -0.0133 0.0143
## # i 14 more rows
```

Dataframe 5: MSE Summary Statistics by Model & Training Proportion

```
df5.mse <- beta1 %>%
  mutate(statistic=mse) %>%
  group_by(train_proportion, model, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df5.mse
```

```
## # A tibble: 24 x 8
##   train_proportion model      regression      mean      sd median      p05      p95
##   <dbl> <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      0.05 gpt-3.5~ Yllm_V      2.27e-4  1.98e-4  1.58e-4  6.59e-5  6.28e-4
## 2      0.05 gpt-3.5~ test_Ytil~ 1.55e-3  5.74e-4  1.49e-3  7.35e-4  2.79e-3
## 3      0.05 gpt-3.5~ train_Yhu~ 1.82e-3  6.41e-4  1.77e-3  9.57e-4  3.15e-3
## 4      0.05 gpt-4o   Yllm_V      1.52e-4  9.73e-5  1.27e-4  5.77e-5  3.75e-4
## 5      0.05 gpt-4o   test_Ytil~ 1.34e-3  5.52e-4  1.26e-3  5.32e-4  2.58e-3
## 6      0.05 gpt-4o   train_Yhu~ 1.83e-3  6.35e-4  1.78e-3  9.86e-4  3.12e-3
## 7      0.1  gpt-3.5~ Yllm_V      2.27e-4  1.97e-4  1.56e-4  6.40e-5  6.51e-4
## 8      0.1  gpt-3.5~ test_Ytil~ 8.17e-4  2.91e-4  7.92e-4  3.91e-4  1.42e-3
## 9      0.1  gpt-3.5~ train_Yhu~ 9.07e-4  3.19e-4  8.96e-4  4.96e-4  1.57e-3
## 10     0.1  gpt-4o   Yllm_V      1.53e-4  9.92e-5  1.30e-4  5.58e-5  3.85e-4
## # i 14 more rows
```

Dataframe 6: Coverage Summary Statistics by Model & Training Proportion

```
df6.coverage <- beta1 %>%
  mutate(statistic=coverage) %>%
  group_by(train_proportion, model, regression) %>%
  summarise(
    mean = mean(statistic),
    sd = sd(statistic),
    median = median(statistic),
    p05 = quantile(statistic, probs[1]),
    p95 = quantile(statistic, probs[2]),
    .groups = "drop")
df6.coverage
```

```
## # A tibble: 24 x 8
##   train_proportion model      regression      mean      sd median      p05      p95
##   <dbl> <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      0.05 gpt-3.5-turbo-0~ Yllm_V      0.761  0.186    0.808  0.372  0.946
## 2      0.05 gpt-3.5-turbo-0~ test_Ytil~ 0.937  0.00803  0.938  0.924  0.95
## 3      0.05 gpt-3.5-turbo-0~ train_Yhu~ 0.946  0.00674  0.946  0.935  0.956
## 4      0.05 gpt-4o   Yllm_V      0.860  0.113    0.917  0.645  0.952
## 5      0.05 gpt-4o   test_Ytil~ 0.937  0.00852  0.938  0.922  0.948
## 6      0.05 gpt-4o   train_Yhu~ 0.945  0.00789  0.944  0.931  0.958
## 7      0.1  gpt-3.5-turbo-0~ Yllm_V      0.760  0.188    0.813  0.369  0.947
## 8      0.1  gpt-3.5-turbo-0~ test_Ytil~ 0.944  0.00716  0.944  0.933  0.955
## 9      0.1  gpt-3.5-turbo-0~ train_Yhu~ 0.948  0.00757  0.948  0.936  0.961
## 10     0.1  gpt-4o   Yllm_V      0.859  0.116    0.916  0.634  0.953
## # i 14 more rows
```

Save Tables

```
# TODO
```

Proxy on the RHS